

Fruit Distribution Acquisition With Multi-Vision for Multi-Arm Harvesting Robots

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Abstract—This study focuses on the precise localization of fruits using multiple stereo visual perception units in a multi-arm collaborative harvesting robot. Fruit localization in overlapping camera fields is challenging in complex and unstructured orchard environments. This is due to the inaccuracies in estimating the fruit centroid of the same target fruit under different camera perspectives, making it difficult to determine whether different centroid positions belong to the same fruit. To address this issue, a novel fruit localization method based on multiple visual units is proposed in this work to reduce mislocalization of fruits in overlapping camera fields. Firstly, a target-based multi-camera extrinsic calibration method is employed to obtain accurate external parameter matrices between multiple cameras and the robot base coordinate system. Secondly, to enhance the recognition and localization accuracy of fruits under a single camera, a multitask fruit detection and segmentation model is used, along with a frustum-based fruit localization method. Finally, to reduce the problem of repeated detections of the same fruit in overlapping camera fields, a filtering method based on fruit surface point cloud matching is proposed. Experimental results demonstrate that the proposed method improves fruit localization accuracy and reduces the probability of repeated false detections. Visualized point cloud information generated by the stereo cameras is provided to showcase the effectiveness of the proposed method.

Keywords—harvesting robot, fruit localization, multiple camera calibration, duplicate target removal

I. INTRODUCTION

With the shortage of labor and the rising costs of manpower, there is a significant increase in the pressure on agricultural labor supply. The development of highly efficient, high-quality, and cost-effective agricultural robots that can replace manual

labor is of great significance for the sustainable development of agricultural production[1].

In recent years, combining modern standardized agricultural practices with multi-arm collaborative harvesting robots has become a consensus in both academia and industry. This approach aims to enhance operational efficiency and has resulted in fruitful research achievements.

Recently, there has been a growing interest in multi-arm harvesting robots [2], [3], [4] aimed at improving operational efficiency in agricultural settings [5]. These robots typically consist of multiple robotic arms, grippers, and vision units, allowing them to perform harvesting operations simultaneously. Significant advancements have been made in this field, including the development of a dual-arm strawberry harvesting robot [6], a 12-arm apple harvesting robot [7], 4-arm kiwifruit harvesting robot[2]. Multi-arm harvesting robots have a larger operational range, and collaborative operations contribute to increased efficiency[8]. These robots typically integrate multiple vision processing units to acquire extensive plant environment information. The aforementioned multi-arm robots are all equipped with visual acquisition devices corresponding to the number of mechanical arms to facilitate gripping tasks. These innovations have contributed to enhancing the efficiency of harvesting robots.

Regarding the planning problem of multi-arm collaborative operations, studies such as [3] propose reinforcement learning-based control schemes for harvesting robot systems to address issues related to overlapping workspace of mechanical arms, avoiding task conflicts and mutual waiting [9], [10].

Meanwhile, there are interferences among the vision acquisition systems closely coordinated with the mechanical arms. In our previous work [3], [11], we presented a multi-arm apple harvesting robot, as shown in Figure 1, where

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visual acquisition units were individually installed on four mechanical arms. Due to the overlapping fields of view of the visual units, the same fruit often appears repetitively in different cameras. By transforming the coordinates using external parameter matrices between cameras, ideally, the position estimation of the same fruit by different cameras should coincide, enabling the elimination of duplicate fruit targets. In practical agricultural operations, due to maintenance requirements, the assembly positions of the robot's stereo cameras often change [12]. Furthermore, with the motion of the mechanical arms, the pose parameters between the cameras frequently experience slow deviations, resulting in changes in their relative positions. As the fruit visual localization algorithm performance is not ideal, there are often varying degrees of deviation in position estimation. These two factors often lead to repeated fruit targets in the collaborative perception of multiple visual units[13], causing the robot to grasp the same fruit multiple times and affecting the efficiency of multi-arm robots. To address this issue, this paper proposes a novel approach for collaborative processing of multiple visual information. Firstly, a multi-stereo camera extrinsic calibration method is employed to rectify the external pose relationships between different cameras. Secondly, to improve the accuracy of fruit recognition and localization by a single visual unit, a multi-task fruit detection and segmentation model is proposed, along with a fruit localization method based on frustum projection, enabling precise fruit localization. Finally, to reduce the issue of repeated detections of the same fruit in overlapping fields of view, a fruit surface point cloud matching filtering method is proposed.

II. METHODOLOGY

A. Multi-arm harvesting robot system

The multi-arm harvesting robot in this work consists of a wheeled mobile platform, mechanical arms, grippers, visual perception system, a fruit conveying system, an illuminating system and a central control module. The robotic arm system includes four units, each of which includes a 3-degree-of-freedom electric-pneumatic hybrid driven Cartesian mechanical arm. The custom-designed 2-DoF gripper has three curved fingers and is able to perform the motions of rotation and grasp to detach the target fruits. The multi-vision system consists of four RGBD cameras that is fixed at the top of third linear module. The RGBD cameras acquire perceptual information of target fruits, and the images are transmitted to the edge computing module to implement detection and segmentation tasks for fruit localization. The control system consists of an industrial personal computer to perform central control and human-machine interaction, two edge computing devices to fulfill image inference task, a CAN bus as the way of communication of control system, and several embedded system circuits.

B. Multi-camera joint calibration

To calibrate the four RGBD cameras, we employ an additional camera $\mathcal{C}_{\text{fixed}}$ that is installed at a fixed position



Fig. 1: The multi-arm apple harvesting robot.

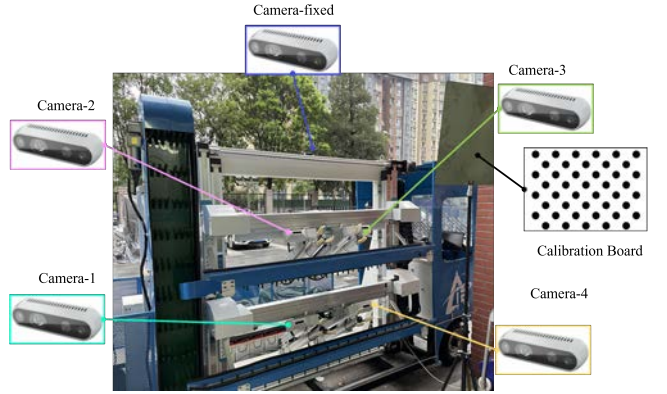


Fig. 2: Four-vision combined calibration configuration.

with respect to the robot base frame $\mathcal{F}_{\text{Base}}$ and the extrinsic parameter matrix $\mathbf{T}_{\text{fixed}}^{\mathcal{C}_{\text{fixed}}}$ is regarded as known. Besides, the four cameras are named by $\mathcal{C}_i$, $i = 1, \dots, 4$.

By analyzing the kinematic relationship of the camera, one may have the relationship between the position vector of feature point P w.r.t. the camera's frame $\mathbf{x}_i^{\mathcal{C}} \in \mathbb{R}^{3 \times 1}$ and the position vector $\mathbf{y} \in \mathbb{R}^{2 \times 1}$ on image plane, satisfying

$$\mathbf{y} = \frac{1}{z} \begin{bmatrix} \Omega_1^T \\ \Omega_2^T \end{bmatrix} \mathbf{x}_i^{\mathcal{C}} \quad (1)$$

where $\Omega_1^T, \Omega_2^T \in \mathbb{R}^{1 \times 3}$ denote 1-st and 2-nd row of intrinsic matrix Ω ; $z \in \mathbb{R}$ is depth. The position vector of P from $\mathcal{C}_{i+1}$ to $\mathcal{C}_i$ satisfies

$$\begin{bmatrix} \mathbf{x}_i^{\mathcal{C}_i} \\ 1 \end{bmatrix} = \mathbf{T}_{\mathcal{C}_{i+1}}^{\mathcal{C}_i} \begin{bmatrix} \mathbf{x}_{i+1}^{\mathcal{C}_{i+1}} \\ 1 \end{bmatrix} \quad (2)$$

where $\mathbf{T}_{\mathcal{C}_{i+1}}^{\mathcal{C}_i} \in SE(3)$ is homogeneous transformation matrix from $\mathcal{C}_{i+1}$ to $\mathcal{C}_i$ and the homogeneous matrices $\mathbf{T}_{\mathcal{C}_{i+1}}^{\mathcal{C}_i}$, $i = 1, \dots, 4$ are extrinsic parameter matrices to be calibrated. According to the above analysis, the relative pose between the fixed camera $\mathcal{C}_{\text{fixed}}$ and the calibration plate satisfies

$$\mathbf{T}_{\text{plate}}^{\mathcal{B}} = \mathbf{T}_{\text{fixed}}^{\mathcal{B}} \prod_{i=1}^3 \mathbf{T}_{\mathcal{C}_{i+1}}^{\mathcal{C}_i} \mathbf{T}_{\text{plate}}^{\mathcal{C}_4} \quad (3)$$

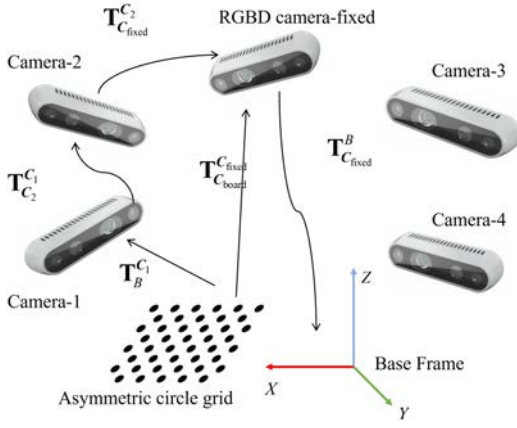


Fig. 3: Multiple cameras joint calibration

In the joint calibration, to calculate the extrinsic matrices of several cameras, the reprojection error model is employed, which can be formulated as follows:

$$\min \sum_{i=1}^n \sum_{j=1}^m \|y_{ij} - \tilde{y}(\Omega, \mathbf{T}, \mathbf{x}_j)\|^2 \quad (4)$$

where n denotes the number of images; m denotes the number of feature points; $\mathbf{x}_j$ represents j -th 3-dimensional space coordinates and $\tilde{y}$ denotes the projection of feature point $\mathbf{x}_j$ on the i -th images. Then we employ Levenberg-Marquardt method to optimize the extrinsic parameter matrices of four cameras.

C. Fruit recognition and localization

1) *Perception Network*: We employed a multi-task deep convolutional neural network that is proposed in our previous work [14]. The network is built upon the YOLOv4 architecture with improvements and consists of a shared encoding network and two decoding networks that perform different tasks. The encoding network comprises a backbone network responsible for feature extraction and a neck network that collects feature maps at different stages. The decoding networks consist of a detection head network, which predicts fruit occlusion types and complete bounding boxes, and an instance segmentation head network that generates pixel masks for the visible parts of fruits as shown in Figure 4.

2) *Training*: The network training is based on a graphic workstation with Dell Precision 7920 Tower (2 Xeon Silver 4214R @ 2.4 GHz CPU cores, 128 GB RAM, and NVIDIA GeForce RTX 3070 (8 GB)). To train the proposed multi-task DCNN, two kinds of datasets were collected: an open-source MinneApple dataset[15] containing 1000 images with over 41,000 labeled instances of apples, and the dataset manually collected in two modern standard orchards in Haidian District and Changping District, Beijing, China, acquired by Realsense D435i RGBD camera, containing 1000 images with over 12k labeled instances and bounding boxes of apples with different kinds of occlusion: non-, leaf-, branch- and fruit-occlusion.

The training and validation sets are distributed into 4:1. In the training stage, we employ a) photometric distortion including random contrast, random saturation, random hue, random brightness, b) random mirror operation, c) random sample crop, d) random flip, e) random expansion to augment the dataset.

3) *Fruit Localization*: Through the object detection branch of a multi-task network, complete bounding boxes of fruits are obtained. By incorporating the camera's internal imaging parameter matrix, a three-dimensional viewing frustum is calculated. Due to the estimation of the two-dimensional image bounding box, which encompasses the complete outer contour of the fruit, even in cases of partial occlusion, the point cloud of the target fruit in the generated RGBD image will also be fully enclosed within the frustum. The symmetric axis of the frustum can be considered as the central line passing through the centroid of the fruit. Subsequently, the segmentation branch of the multi-task network enables the extraction of fruit masks. Similarly, based on the camera's internal parameters and aligned depth map information, a three-dimensional point cloud of the mask, specifically the fruit surface point cloud, can be generated. By applying DBSCAN clustering analysis, the cluster centroids of the fruit surface point cloud can be extracted. Compared to clustering methods such as K-Means and BIRCH, which are generally suitable for convex sample sets, DBSCAN can be applied to both convex and non-convex sample sets. Utilizing DBSCAN helps filter out noise points within the frustum, including outliers caused by occlusions and excessively smoothed disturbance surfaces that are not part of the fruit surface point cloud. After obtaining reference points on the fruit surface, the radius of the target fruit can be further determined by utilizing the camera's intrinsic parameters and the two-dimensional bounding box information. Subsequently, by combining this radius information with the known central line of the fruit, the centroid position of the fruit can be calculated.

D. Multi-vision perception information fusion

Once the recognition and localization tasks are finished, it is necessary to fuse the fruit position information obtained from different visual units to form a global fruit distribution map. Due to the fact that different cameras localize the same fruit from different perspectives, there are differences in the number of point clouds on the fruit surface and the degree of occlusion captured from different viewpoints.

Step 1: Utilizing camera extrinsic parameters and the respective field of view, objects detected by each camera that do not reside within the overlapping field of view are transferred from the original list to the global fruit list.

Step 2: For each object detected within the overlapping regions by i -th camera, if the n -th target, $O_{i,n}$, is located in adjacent overlapping regions between cameras, the position $\hat{P}_i$ of the center of the detection box P_i in i -th camera is transformed using the intrinsic and extrinsic matrices of i -th camera and the adjacent camera. Then iterate through the objects in the overlapping region of the adjacent camera. If the

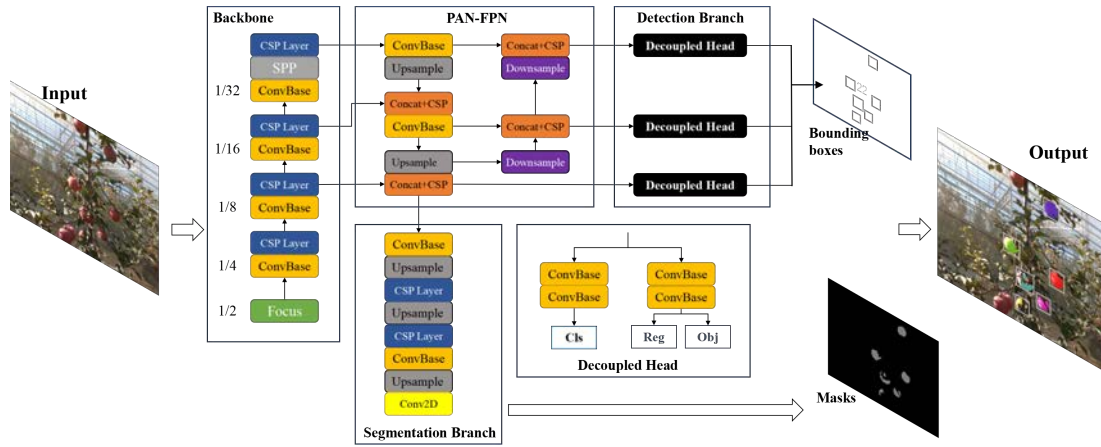


Fig. 4: Multi-task network for fruit identification.

Euclidean distance between the projected positions of $O_{j,n}$ and $O_{i,n}$ in the adjacent camera image is less than a threshold λ (i.e., $\|P_{j,n} - \hat{P}_i\|^2 < \lambda$), it indicates that they represent the same target. Consequently, the target is removed from the fruit lists of both i -th camera and the adjacent camera and added to the global fruit list. If the distance exceeds the threshold, $O_{j,n}$ is considered a distinct object and is retained in the original list.

Step 3: If the target $O_{i,n}$ is present within the overlapping field of view of all four cameras, apply the aforementioned procedure iteratively to determine if it represents the same object in the other cameras, excluding i -th camera. If $O_{i,n}$ appears in the other cameras more than once, it is eliminated from the fruit lists of i -th camera and the other cameras, and subsequently appended to the global fruit list. If $O_{i,n}$ does not appear in the other cameras, it is classified as a suspicious fruit and retained in the original list.

Step 4: Repeat the above process sequentially for the remaining cameras until all objects detected within their respective overlapping field of view have undergone the uniqueness verification.

Step 5: Transmit the generated global fruit list to the task planner.

Step 6: Once the global fruit list is picked, the robotic arms return back to the observing site to start to acquire fruit distribution again and repeat the above five steps.

III. EXPERIMENTAL RESULTS

A. Multi-camera calibration

The main control is implemented by a industrial personal computer with CPU Intel Core™ i7-1165G7 processor and GPU Nvidia Geforce RTX 2060. To calibrate multi-camera extrinsic matrices, a checkerboard is expected to stay in the common FoV of multiple cameras, at least two cameras. It is noteworthy that calibrating all cameras relative to the fixed camera is the fastest calibration method, as shown in Fig.5. This means that all cameras can simultaneously detect the calibration object. For four cameras, multiple calibration steps are still required to calculate extrinsic matrix between each

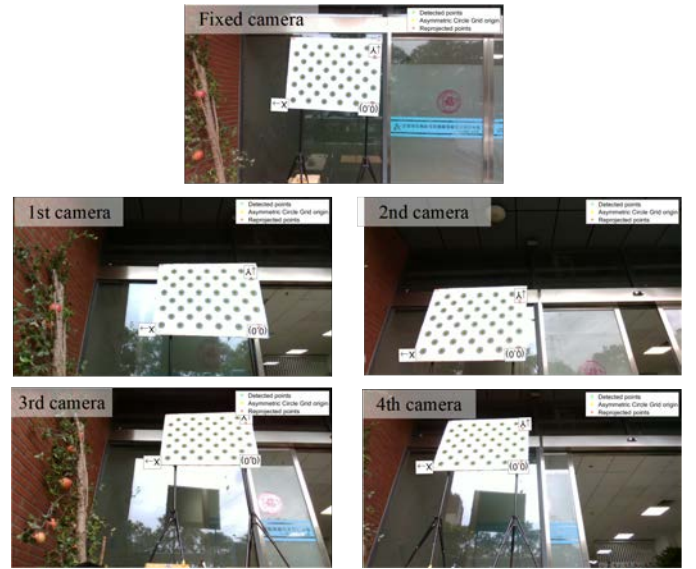


Fig. 5: Multi-camera calibration.

camera and the fixed camera step by step. Every step generates a matrix denoting the transformation from current camera frame to the fixed camera frame.

B. Fruit Localization

To evaluate the performance of the proposed fruit localization method at various distances, a comprehensive series of experiments was conducted. Three sets of RGBD images were acquired at distances approximately 500 mm, 800 mm, and 1000 mm from the fruit working surface. As a benchmark for comparative analysis, the widely adopted object detection-based fruit localization method, abbreviated as "bbx. mtd." for brevity, was employed. The experimental findings are summarized in Table 2.3.

The results reveal that, compared to the reference method, the proposed approach yields significant reductions in the median error, average error, and standard error associated with fruit centroid estimation. Specifically, in Group 1, the proposed

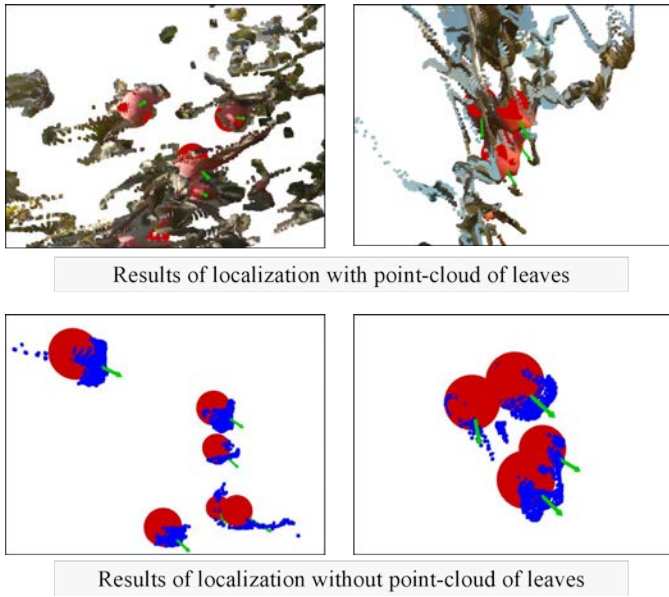


Fig. 6: Fruit localization results by the proposed method.

method demonstrates respective reductions of 67%, 50%, and 10% in these error metrics. Across all three experimental groups, the reductions amount to 59%, 43%, and 9%, respectively.

With respect to fruit diameter measurement, the proposed method exhibits substantial improvements over the reference method. In group 1, the proposed method achieves reductions of 75%, 80%, and 96% in the median error, average error, and standard error, respectively. The overall analysis across the three groups indicates reductions of 70%, 70%, and 78% for these error metrics.

These results underscore the effectiveness of the proposed method in mitigating errors in both centroid position estimation and fruit diameter measurement.

C. Fruit distribution information fusion

To evaluate the fusion performance of the proposed method on large-scale fruit working surfaces, extensive experiments were conducted in an actual orchard. Ten representative working points were carefully selected, each exhibiting varying quantities and distributions of apple fruits. RGBD images were captured from four cameras at each working point. The experimental results, depicted in Figures 7-10 demonstrate the effectiveness of the proposed approach.

Figure 7 illustrates the overlapping field of view between camera-3 and camera-4. The green dots represent the bounding box centers of identified fruit targets within this region, while the blue points show the projections of these targets in camera -4 onto camera-3. It can be seen that the projection points are at an ideal distance from their corresponding center points under camera-3, verifying that these pairs of points represent the same targets. The visual representation demonstrates accurate recognition of the same fruit target within the overlapping view field.



Fig. 7: Detections and projections of fruits in the overlapping field of view between camera-4 and camera-3.

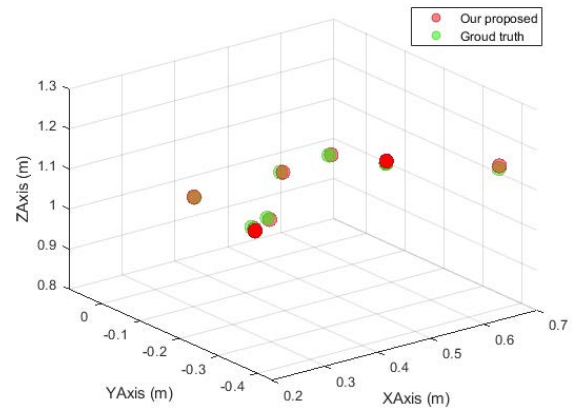


Fig. 8: Results of fruit distribution by the fusion of multi camera views.

Figure 8 showcases the successful fusion of fruit distribution at the third working point. The green spheres represent the actual fruit positions obtained through manual measurements, assisted by laser radar. The red spheres represent the fruit distribution obtained using the proposed method. This figure shows that the red spheres are in close agreement with the green spheres, demonstrating the precise localization and fusion of fruit distribution achieved by the proposed approach.

Figures 9 and 10 present cases where the proposed method encountered challenges. In Figure 9, due to factors like image detection inaccuracies, false positives, and fruit occlusions, there are significant image differences of fruit targets within the overlapping field of view between adjacent cameras, resulting in large deviations in target projection points and misjudgments of non-identical targets. Figure 10 illustrates missed detections that occur when such misjudgments happen.

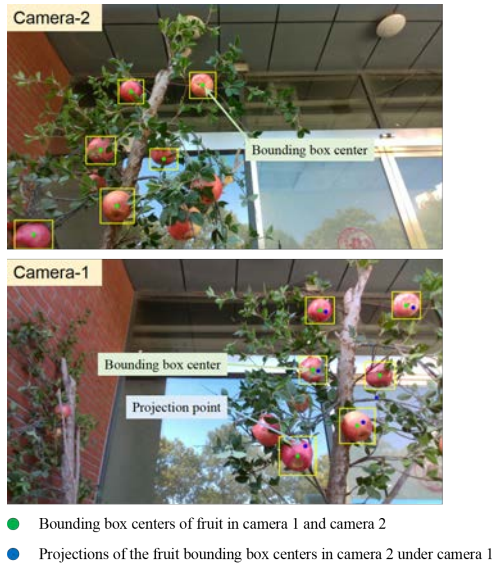


Fig. 9: Detections and projections of fruits in the overlapping field of view between camera-2 and camera-1.

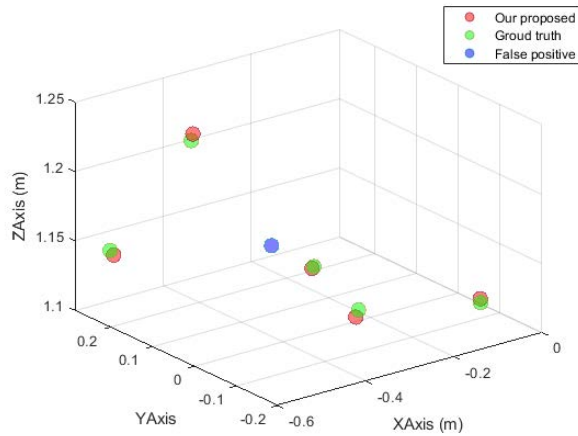


Fig. 10: Results of fruit distribution by the fusion of multi camera views with a false positive.

During the experiment, adjusting the image distance threshold λ helped relax the criteria for non-identical targets and reduce missed detections to some extent. However, caution should be exercised as this approach may mistakenly identify close non-identical targets as identical.

These findings highlight the effectiveness of the proposed method in reducing errors in fruit centroid position estimation and fruit diameter measurement. The fruit centroid position and diameter measurement errors were on average 9.2 mm and 8.6 mm, respectively, compared to 10.2 mm and 9.7 mm for the comparison in [14]. The results contribute to advancing fruit distribution information acquisition and validating the potential of the proposed method in enhancing the performance of fruit harvesting robotic systems.

IV. CONCLUSION

This study investigates the problem of acquiring large-scale fruit distribution information through multi-vision techniques. Firstly, a deep-learning-based network is proposed for the simultaneous detection and segmentation of fruits. The positional accuracy is improved through the utilization of the DBSCAN point-cloud-processing method. Secondly, an information fusion method is presented for fruit distribution by employing joint calibration of multiple vision units and introducing a fusion algorithm to determine the identity of fruits across different cameras.

Lastly, a complete series of experiments was conducted to evaluate the effectiveness of the proposed method. The experimental results demonstrate that the proposed approach effectively integrates fruit distribution information from various vision units, resulting in a reduction of 20% in the probability of false positives and the elimination of duplicated targets during the fusion process.

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